

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2459949.776 +/- 0.079 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.152 +/- 0.051
Transit depth (Rp/Rs)^2	2.3 +/- 1.55 [%]
Orbital Inclination (inc)	80.74 +/- 2.89
Airmass coefficient 1 (a1)	2.3711 +/- 0.0014
Airmass coefficient 2 (a2)	-0.63 +/- 1.33
Scatter in the residuals of the lightcurve fit is	0.053 %
Best Comparison Star	#3 - [71, 341]
Optimal Aperture	45.71
Optimal Annulus	76.18
Transit Duration (day)	0.005 +/- 0.013

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.152 ± 0.051 vs 0.1404 (deviation 0.23σ)
Duration fit vs published	0.005 d vs 0.0754 d (deviation 5.42σ)
Mid-time O–C	-2779.8 min
Depth SNR	3.0
Residual scatter	0.053 %

Light curve

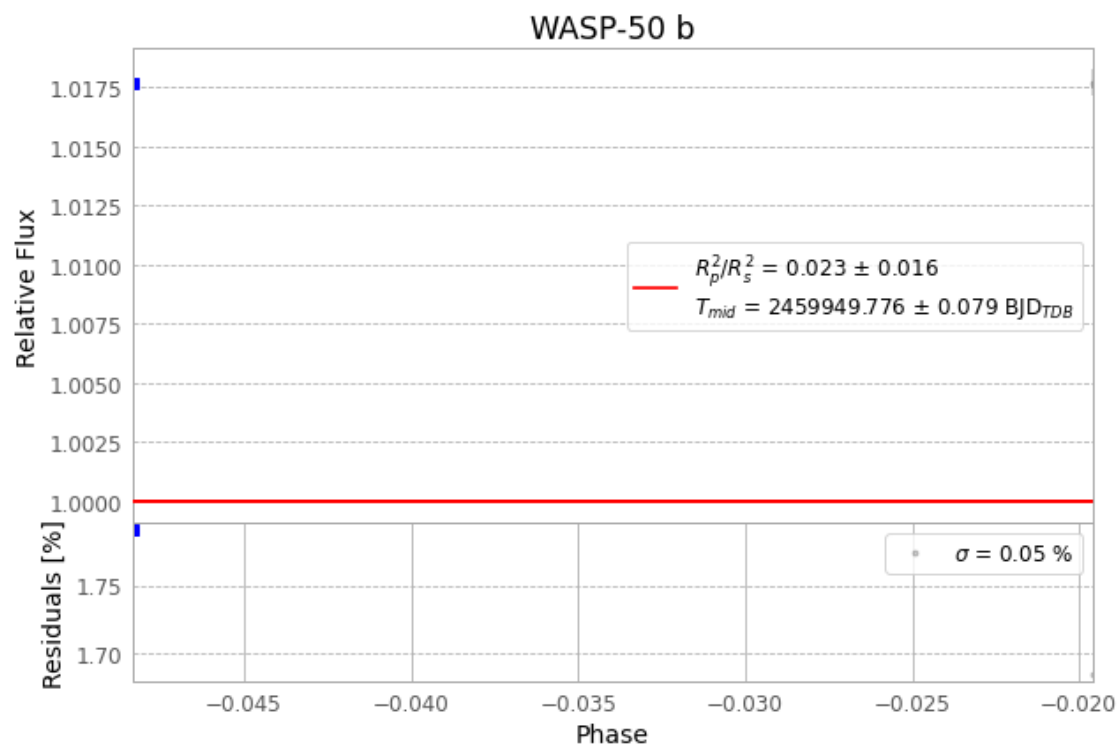


Figure 1 — FinalLightCurve_WASP-50 b_2023-01-06.png (SHA-256 165e2a3115eb49ad...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [517, 244], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 d12053ff0cb507ee...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

77 FITS frames (51 MB), WASP-50230107030018.FITS → WASP-50230107064815.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-50-230107.log (66dc063e56a2...), FinalLightCurve_WASP-50 b_2023-01-06.png (165e2a3115eb...), FinalLightCurve_WASP-50 b_2023-01-06.csv (3708700a3b29...)

Compute resources

Wall time 461.0 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-28 04:44Z.